

Caught in a trap: DNA contamination in tsetse xenomonitoring can lead to over-estimates of *Trypanosoma brucei* infection.

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Tsetse flies (*Glossina* sp.) are vectors of *Trypanosoma brucei* subspecies that cause human African trypanosomiasis (HAT). Capturing and screening tsetse is critical for HAT surveillance. Classically, tsetse have been microscopically analysed to identify trypanosomes, but this is increasingly replaced with molecular xenomonitoring. Nonetheless, sensitive *T. brucei*-detection assays, such as TBR-PCR, are vulnerable to DNA cross-contamination. This may occur at capture, when often multiple live tsetse are retained temporarily in the cage of a trap. This study set out to determine whether infected tsetse can contaminate naïve tsetse with *T. brucei* DNA via faeces when co-housed. Insectary-reared teneral *G. morsitans morsitans* were fed an infectious *T. b. brucei*-spiked bloodmeal. At 19 days post-infection, infected and naïve tsetse were caged together in the following ratios: (T1) 9:3, (T2) 6:6 (T3) 1:11 and a control (C0) 0:12 in triplicate. Following 24-hour incubation, DNA was extracted from each fly and screened for parasite DNA presence using PCR and qPCR. All insectary-reared infected flies were positive for *T. brucei* DNA using TBR-qPCR. However, naïve tsetse also tested positive. Even at a ratio of 1 infected to 11 naïve flies, 91% of naïve tsetse gave positive TBR-qPCR results. Furthermore, the quantity of *T. brucei* DNA detected in naïve tsetse was significantly correlated with cage infection ratio. With evidence of cross-contamination, field-caught tsetse from Tanzania were then assessed using the same screening protocol. End-point TBR-PCR predicted a sample population prevalence of 24.8%. Using qPCR and Cq cut-offs optimised on insectary-reared flies, we estimated that prevalence was 0.5% (95% confidence interval [0.36, 0.73]). Our results show that infected tsetse can contaminate naïve flies with *T. brucei* DNA when co-caged, and that the level of contamination can be extensive. Whilst simple PCR may overestimate infection prevalence, quantitative PCR offers a means of eliminating false positives.

TFPP: an SVM-based tool for recognizing flagellar proteins in *Trypanosoma brucei*.

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Trypanosoma brucei is a unicellular flagellated eukaryotic parasite that causes African trypanosomiasis in human and domestic animals with devastating health and economic consequences. Recent studies have revealed the important roles of the single flagellum of *T. brucei* in many aspects, especially that the flagellar motility is required for the viability of the bloodstream form *T. brucei*, suggesting that impairment of the flagellar function may provide a promising cure for African sleeping sickness. Knowing the flagellum proteome is crucial to study the molecular mechanism of the flagellar functions. Here we present a novel computational method for identifying flagellar proteins in *T. brucei*, called trypanosome flagellar protein predictor (TFPP). TFPP was developed based on a list of selected discriminating features derived from protein sequences, and could predict flagellar proteins with ~92% specificity at a ~84% sensitivity rate. Applied to the whole *T. brucei* proteome, TFPP reveals 811 more flagellar proteins with high confidence, suggesting that the flagellar proteome covers ~10% of the whole proteome. Comparison of the expression profiles of the whole *T. brucei* proteome at three typical life cycle stages found that ~45% of the flagellar proteins were significantly changed in expression levels between the three life cycle stages, indicating life cycle stage-specific regulation of flagellar functions in *T. brucei*. Overall, our study demonstrated that TFPP is highly effective in identifying flagellar proteins and could provide opportunities to study the trypanosome flagellar proteome systematically. Furthermore, the web server for TFPP can be freely accessed at <http://wukong.tongji.edu.cn/tfpp>.

Extracellular-signal regulated kinase 8 of *Trypanosoma brucei* uniquely phosphorylates its proliferating cell nuclear antigen homolog and reveals exploitable properties.

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The *Trypanosoma brucei* subspecies *T. brucei gambiense* and *T. brucei rhodesiense* are vector-borne pathogens that cause sleeping sickness also known as Human African Trypanosomiasis (HAT), which is fatal if left untreated. The drugs that treat HAT are ineffective and cause toxic side effects. One strategy for identifying safer and more effective HAT drugs is to therapeutically exploit essential gene targets in *T. brucei*. Genes that make up a basic mitogen-activated protein kinase (MAPK) network are present in *T. brucei*. Tb927.10.5140 encodes an essential MAPK that is homologous to the human extracellular-signal regulated kinase 8 (HsERK8) which forms a tight complex with the replication factor proliferating cell nuclear antigen (PCNA) to stabilize intracellular PCNA levels. Here we demonstrate that (TbPCNA) is uniquely phospho-rylated on serine (S) and threonine (T) residues in *T. brucei* and that TbERK8 phosphorylates TbPCNA at each of these residues. The ability of an ERK8 homolog to phosphorylate a PCNA homolog is a novel biochemical property that is first demonstrated here in *T. brucei* and may be unique to this pathogen. We demonstrate that the potent HsERK8 inhibitor Ro318220, has an IC₅₀ for TbERK8 that is several hundred times higher than its reported IC₅₀ for HsERK8. This indicated that the active sites of TbERK8 and HsERK8 can be selectively inhibited, which provides a rational basis for discovering inhibitors that specifically target this essential parasite MAPK to kill the parasite.

Current chemotherapy of human African trypanosomiasis.

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Human African trypanosomiasis is a fatal disease caused by *Trypanosoma brucei gambiense* and *Trypanosoma brucei rhodesiense* that has re-emerged in recent years. However, very little progress has been made in the development of new drugs against this disease. Most drugs still in use were developed one or more decades ago, and are generally toxic and of limited effectiveness. The most recently introduced compound, eflornithine, is only useful against sleeping sickness caused by *T. b. gambiense*, and is prohibitively expensive for the African developing countries. We present here an overview of today's approved and clinically used drugs against this disease.

Novel 4-[4-(4-methylpiperazin-1-yl)phenyl]-6-arylpyrimidine derivatives and their antitrypanosomal activities against *T.brucei*.

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Human African trypanosomiasis, or sleeping sickness, is a neglected tropical disease caused by *Trypanosoma brucei rhodesiense* and *Trypanosoma brucei gambiense* and is invariably fatal unless treated. Current therapies present limitations in their application, parasite resistance, or require further clinical investigation for wider use. Our work, informed by previous findings, presents novel 4-[4-(4-methylpiperazin-1-yl)phenyl]-6-arylpyrimidine derivatives with promising antitrypanosomal activity. In particular, 32 exhibits an in vitro EC₅₀ value of 0.5 μ M against *Trypanosoma brucei*

rhodesiense, and analogues 29, 30 and 33 show antitrypanosomal activities in the $<1 \mu\text{M}$ range. We have demonstrated that substituted 4-[4-(4-methylpiperazin-1-yl)phenyl]-6-arylpyrimidines present promising antitrypanosomal hit molecules with potential for further preclinical development.

Activity of indenoisoquinolines against African trypanosomes.

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African trypanosomiasis (sleeping sickness), caused by protozoan *Trypanosoma brucei* species, is a debilitating disease that is lethal if untreated. Available drugs are antiquated, toxic, and compromised by emerging resistance. The indenoisoquinolines are a class of noncamptothecin topoisomerase IB poisons that are under development as anticancer agents. We tested a variety of indenoisoquinolines for their ability to kill *T. brucei*. Indenoisoquinolines proved trypanocidal at submicromolar concentrations in vitro. Structure-activity analysis yielded motifs that enhanced potency, including alkylamino substitutions on N-6, methoxy groups on C-2 and C-3, and a methylenedioxy bridge between C-8 and C-9. Detailed analysis of eight water-soluble indenoisoquinolines demonstrated that in trypanosomes the compounds inhibited DNA synthesis and acted as topoisomerase poisons. Testing these compounds on L1210 mouse leukemia cells revealed that all eight were more effective against trypanosomes than against mammalian cells. In preliminary in vivo experiments one compound delayed parasitemia and extended survival in mice subjected to a lethal trypanosome challenge. The indenoisoquinolines provide a promising lead for the development of drugs against sleeping sickness.

Cell Cycle Inhibition To Treat Sleeping Sickness.

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African trypanosomiasis is caused by infection with the protozoan parasite *Trypanosoma brucei*. During infection, this pathogen divides rapidly to high density in the bloodstream of its mammalian host in a manner similar to that of leukemia. Like all eukaryotes, *T. brucei* has a cell cycle involving the de novo synthesis of DNA regulated by ribonucleotide reductase (RNR), which catalyzes the conversion of ribonucleotides into their deoxy form. As an essential enzyme for the cell cycle, RNR is a common target for cancer chemotherapy. We hypothesized that inhibition of RNR by genetic or pharmacological means would impair parasite growth in vitro and prolong the survival of infected animals. Our results demonstrate that RNR inhibition is highly effective in suppressing parasite growth both in vitro and in vivo. These results support drug discovery efforts targeting the cell cycle, not only for African trypanosomiasis but possibly also for other infections by eukaryotic pathogens. **IMPORTANCE** The development of drugs to treat infections with eukaryotic pathogens is challenging because many key virulence factors have closely related homologues in humans. Drug toxicity greatly limits these development efforts. For pathogens that replicate at a high rate, especially in the blood, an alternative approach is to target the cell cycle directly, much as is done to treat some hematologic malignancies. The results presented here indicate that targeting the cell cycle via inhibition of ribonucleotide reductase is effective at killing trypanosomes and prolonging the survival of infected animals.

Specific Cell Targeting Therapy Bypasses Drug Resistance Mechanisms in African Trypanosomiasis.

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African trypanosomiasis is a deadly neglected disease caused by the extracellular parasite *Trypanosoma brucei*. Current therapies are characterized by high drug toxicity and increasing drug resistance mainly associated with loss-of-function mutations in the transporters involved in drug import. The introduction of new antiparasitic drugs into therapeutic use is a slow and expensive process. In contrast, specific targeting of existing drugs could represent a more rapid and cost-effective approach for neglected disease treatment, impacting through reduced systemic toxicity and circumventing resistance acquired through impaired compound uptake. We have generated nanoparticles of chitosan loaded with the trypanocidal drug pentamidine and coated by a single domain nanobody that specifically targets the surface of African trypanosomes. Once loaded into this nanocarrier, pentamidine enters trypanosomes through endocytosis instead of via classical cell surface transporters. The curative dose of pentamidine-loaded nanobody-chitosan nanoparticles was 100-fold lower than pentamidine alone in a murine model of acute African trypanosomiasis. Crucially, this new formulation displayed undiminished in vitro and in vivo activity against a trypanosome cell line resistant to pentamidine as a result of mutations in the surface transporter aquaglyceroporin 2. We conclude that this new drug delivery system increases drug efficacy and has the ability to overcome resistance to some anti-protozoal drugs.

Bistacrines as potential antitrypanosomal agents.

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Human African Trypanosomiasis (HAT) is caused by two subspecies of the genus *Trypanosoma*, namely *Trypanosoma brucei rhodesiense* and *Trypanosoma brucei gambiense*. The disease is fatal if left untreated and therapy is limited due to only five non-adequate drugs currently available. In preliminary studies, dimeric tacrine derivatives were found to inhibit parasite growth with IC50-values in the nanomolar concentration range. This prompted the synthesis of a small, but smart library of monomeric and dimeric tacrine-type compounds and their evaluation of antiprotozoal activity. Rhodesain, a lysosomal cathepsin-L like cysteine protease of *T. brucei rhodesiense* is essential for parasite survival and likely target of the tacrine derivatives. In addition, the inhibition of trypanothione reductase by bistacrines was found. This flavoprotein oxidoreductase is the main defense against oxidative stress in the thiol redox system unique for protozoa.

The burden of human African trypanosomiasis.

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Human African trypanosomiasis (HAT, or sleeping sickness) is a protozoan parasitic infection caused by *Trypanosoma brucei rhodesiense* or *Trypanosoma brucei gambiense*. These are neglected tropical diseases, and T.b. *rhodesiense* HAT is a zoonosis. We review current knowledge on the burden of HAT in sub-Saharan Africa, with an emphasis on the disability-adjusted life year (DALY), data sources, and methodological issues relating to the use of this metric for assessing the burden of this disease. We highlight areas where data are lacking to properly quantify the impact of these diseases, mainly relating

to quantifying under-reporting and disability associated with infection, and challenge the HAT research community to tackle the neglect in data gathering to enable better evidence-based assessments of burden using DALYs or other appropriate measures.
